

Single Correct Answer Type

- 1) The sum and difference of two perpendicular vectors of equal length are
 (a) also perpendicular and of equal length
 (b) also perpendicular and of different lengths
 (c) of equal length and have an obtuse angle between them
 (d) of equal length and have an acute angle between them

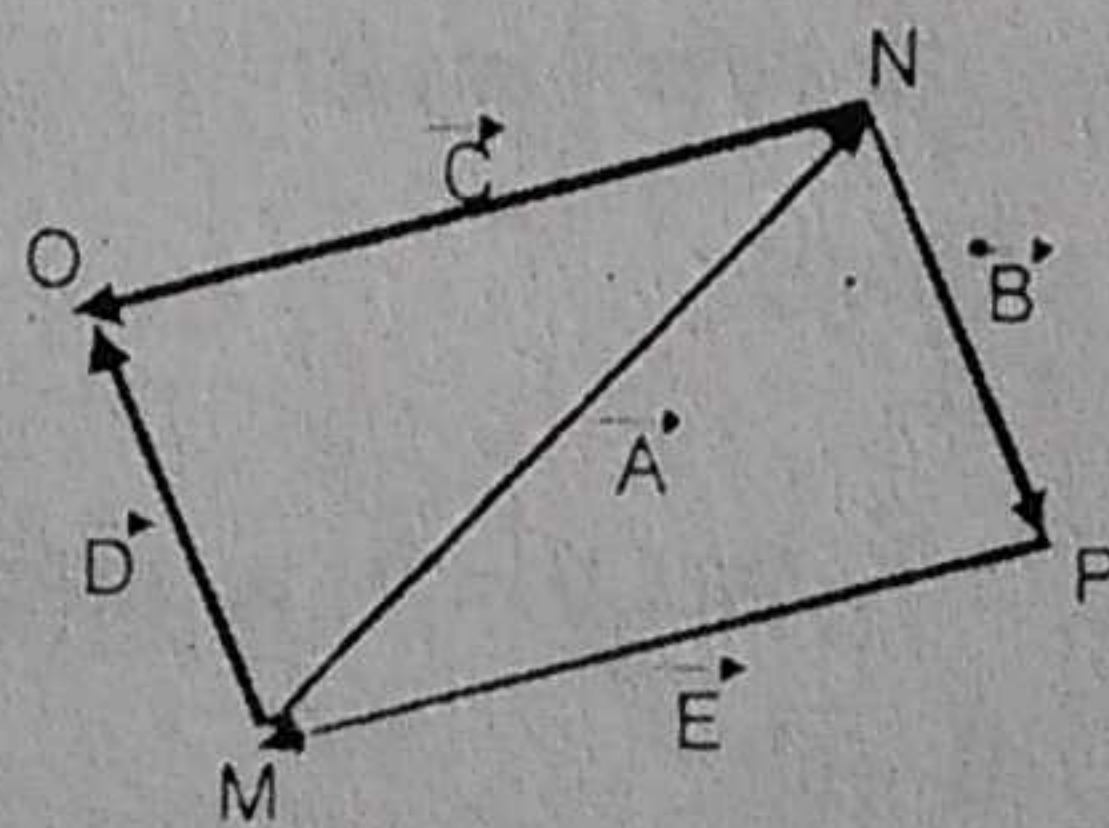
- 2) The minimum number of vectors having different planes which can be added to give zero resultant is
 (a) 2 (b) 3 (c) 4 (d) 5

- 3) Vectors perpendicular to $\hat{i} + \hat{j} + \hat{k}$ is

- (a) $\hat{i} - \hat{j} + \hat{k}$ (b) $\hat{i} - \hat{j} - \hat{k}$ (c) $-\hat{i} - \hat{j} - \hat{k}$ (d) $3\hat{i} + 2\hat{j} - 5\hat{k}$

- 4) From fig. the correct relation is

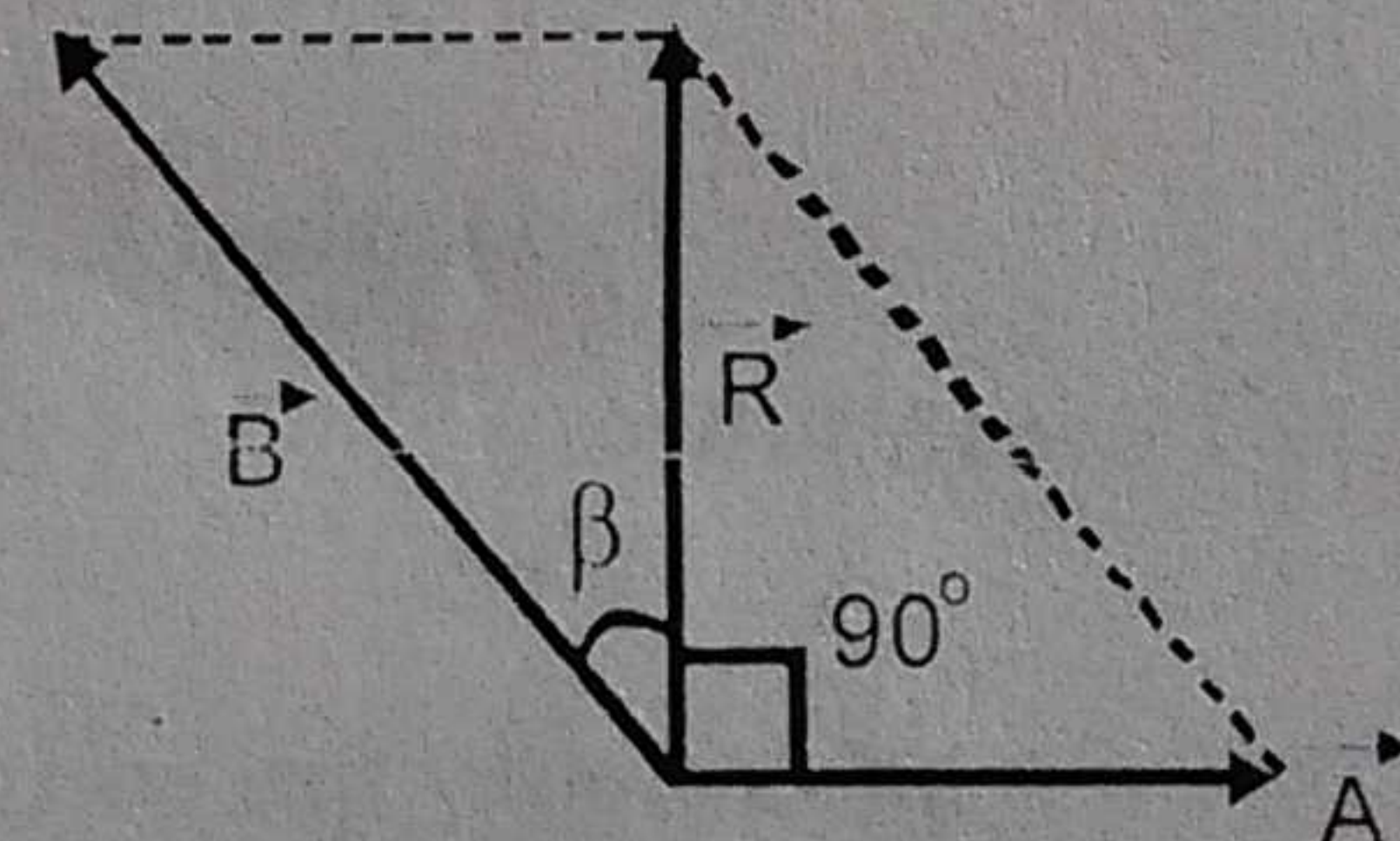
- (a) $\vec{A} + \vec{B} + \vec{E} = \vec{0}$
 (b) $\vec{C} - \vec{D} = -\vec{A}$
 (c) $\vec{B} + \vec{E} - \vec{C} = -\vec{D}$
 (d) All of these



- 5) Out of the following set of forces, the resultant of which cannot be zero
 (a) 10, 10, 10 (b) 10, 10, 20 (c) 10, 20, 20 (d) 10, 20, 40

- 6) The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to the vector $\vec{A}$ and its magnitude is equal to half of the magnitude of vector $\vec{B}$.
 The angle between $\vec{A}$ and $\vec{B}$ is

- (a) 120°
 (b) 150°
 (c) 135°
 (d) None of these

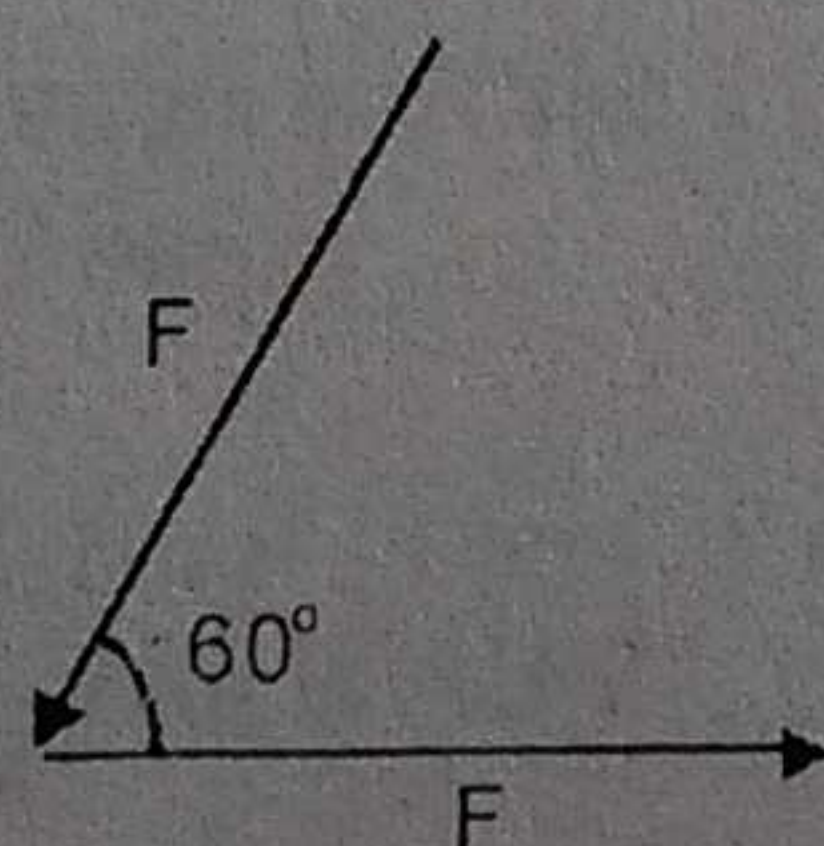


- 7) The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{a}$ and $\vec{b}$ is 3 : 1. Now, $|\vec{a}|$ is equal to

- (a) $|\vec{b}|$ (b) $2|\vec{b}|$ (c) $3|\vec{b}|$ (d) $4|\vec{b}|$

- 8) Two forces, each equal to F, act as shown in fig. Their resultant is

- (a) $F/2$
 (b) F
 (c) $\sqrt{3}F$
 (d) $\sqrt{5}F$



$$\begin{aligned} a + b &= 3a - 3b \\ 0 &= 2a - 4b \\ 0 &= a - 2b \\ \boxed{a = 2b} \end{aligned}$$

- 9) Vector $\vec{A}$ is 2 cm long and is 60° above the x-axis in the first quadrant. Vector $\vec{B}$ is 2 cm long and is 60° below the x-axis in the fourth quadrant. The sum $\vec{A} + \vec{B}$ is a vector of magnitude
 (a) 2 cm along + y-axis (b) 2 cm along + x-axis (c) 2 cm along - x axis (d) 2 cm along - x-axis

- 10) What is the angle between two vector forces of equal magnitude such that the resultant is one-third as much as either of the original forces?

- (a) $\cos^{-1}\left(-\frac{17}{18}\right)$ (b) $\cos^{-1}\left(\frac{1}{3}\right)$ (c) 45° (d) 120°

- 11) The angle between $\vec{A} + \vec{B}$ and $\vec{A} \times \vec{B}$ is
 (a) 0 (b) $\pi/4$ (c) $\pi/2$ (d) π
- 12) The projection of a vector $\vec{r} = 3\hat{i} + \hat{j} + 2\hat{k}$ on the $x - y$ plane has magnitude
 (a) 3 (b) 4 (c) $\sqrt{14}$ (d) $\sqrt{10}$
- 13) If $|\vec{A} + \vec{B}| = |\vec{A}| = |\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ is
 (a) 120° (b) 60° (c) 90° (d) 0°
- 14) If vectors $\vec{A} = \hat{i} + 2\hat{j} + 4\hat{k}$ and $\vec{B} = 5\hat{i}$ represent the two sides of a triangle, then the third side of the triangle can have length equal to
 (a) 6 (b) $\sqrt{56}$ (c) Both the above (d) None of the above
- 15) Given $|\vec{A}_1| = 2$, $|\vec{A}_2| = 3$ and $|\vec{A}_1 + \vec{A}_2| = 3$. Find the value of $(\vec{A}_1 + 2\vec{A}_2) \cdot (3\vec{A}_1 - 4\vec{A}_2)$
 (a) 64 (b) -64 (c) 62 (d) 61
- 16) Three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ satisfy the relation $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \cdot \vec{C} = 0$. The vector $\vec{A}$ is parallel to
 (a) $\vec{B}$ (b) $\vec{C}$ (c) $\vec{B} \cdot \vec{C}$ (d) $\vec{B} \times \vec{C}$
- 17) If $\vec{A} = \vec{B} + \vec{C}$, and the magnitudes of $\vec{A}$, $\vec{B}$, $\vec{C}$ are 5, 4 and 3 units, then angle between $\vec{A}$ and $\vec{C}$ is
 (a) $\cos^{-1}\left(\frac{3}{5}\right)$ (b) $\cos^{-1}\left(\frac{4}{5}\right)$ (c) $\sin^{-1}\left(\frac{3}{4}\right)$ (d) $\frac{\pi}{2}$
- 18) Given: $\vec{A} = A \cos \theta \hat{i} + A \sin \theta \hat{j}$. A vector $\vec{B}$ which is perpendicular to $\vec{A}$ is given by
 (a) $B \cos \theta \hat{i} - B \sin \theta \hat{j}$ (b) $B \sin \theta \hat{i} - B \cos \theta \hat{j}$ (c) $B \cos \theta \hat{i} + B \sin \theta \hat{j}$ (d) $B \sin \theta \hat{i} + B \cos \theta \hat{j}$
- 19) The angle which the vector $\vec{A} = 2\hat{i} + 3\hat{j}$ makes with y-axis, where $\hat{i}$ and $\hat{j}$ are unit vectors along x- and y- axes, respectively, is
 (a) $\cos^{-1}(3/5)$ (b) $\cos^{-1}(2/3)$ (c) $\tan^{-1}(2/3)$ (d) $\sin^{-1}(2/3)$
- 20) Given: $\vec{P} = 3\hat{i} - 4\hat{j}$. Which of the following is perpendicular to $\vec{P}$?
 (a) $3\hat{i}$ (b) $4\hat{j}$ (c) $4\hat{i} + 3\hat{j}$ (d) $4\hat{i} - 3\hat{j}$
- 21) In going from one city to another, a car travels 75 km north, 60 km north west and 20 km east. The magnitude of displacement between the two cities is (Take $1/\sqrt{2} = 0.7$)
 (a) 170 km (b) 137 km (c) 119 km (d) 140 km
- 22) What is the angle between $\vec{A}$ and $\vec{B}$. If $\vec{A}$ and $\vec{B}$ are the adjacent sides of parallelogram drawn from a common point and the area of the parallelogram is $AB/2$?
 (a) 15° (b) 30° (c) 45° (d) 60°
- 23) Two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$. What is the angle between $\vec{a}$ and $\vec{b}$?
 (a) 0° (b) 90° (c) 60° (d) 180°
- 24) Given: $\vec{A} = 4\hat{i} + 6\hat{j}$ and $\vec{B} = 2\hat{i} + 3\hat{j}$. Which of the following is correct?
 (a) $\vec{A} \times \vec{B} = 0$ (b) $\vec{A} \cdot \vec{B} = 24$ (c) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$ (d) $\vec{A}$ and $\vec{B}$ are anti parallel

- 25) Given $\vec{A} = 2\hat{i} + p\hat{j} + q\hat{k}$ and $\vec{B} = 5\hat{i} + 7\hat{j} + 3\hat{k}$. If $\vec{A} \parallel \vec{B}$. Then the values of p and q are, respectively
 (a) $\frac{14}{5}$ and $\frac{6}{5}$ (b) $\frac{14}{3}$ and $\frac{6}{5}$ (c) $\frac{6}{5}$ and $\frac{1}{3}$ (d) $\frac{3}{4}$ and $\frac{1}{4}$
- 26) If $\vec{A}$ is perpendicular to $\vec{B}$, then $\vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0$
 (a) $\vec{A} \times \vec{B} = 0$ (b) $\vec{A} \cdot [\vec{A} + \vec{B}] = A^2$ (c) $\vec{A} \cdot \vec{B} = AB$ (d) $\vec{A} \cdot [\vec{A} + \vec{B}] = A^2 = AB$
- 27) If the angle between vectors $\vec{a}$ and $\vec{b}$ is an acute angle, then the difference $\vec{a} - \vec{b}$ is
 (a) the major diagonal of the parallelogram (b) the minor diagonal of the parallelogram
 (c) any of the above (d) none of the above
- 28) Given the $\vec{A} + \vec{B} = \vec{C}$. If $|\vec{A}| = 4$, $|\vec{B}| = 5$ and $|\vec{C}| = \sqrt{61}$. The angle between $\vec{A}$ and $\vec{B}$ is
 (a) 30° (b) 60° (c) 90° (d) 120°
- 29) Given vector $\vec{A} = 2\hat{i} + 3\hat{j}$, the angle between $\vec{A}$ and y-axis is
 (a) $\tan^{-1}(3/2)$ (b) $\tan^{-1}(2/3)$ (c) $\sin^{-1}(2/3)$ (d) $\cos^{-1}(2/3)$
- 30) If $\vec{b} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = \hat{i} - \hat{j}$, the vector having the same magnitude as that of $\vec{b}$ and parallel to $\vec{a}$ is vector $= |\vec{b}| \frac{\vec{a}}{|\vec{a}|}$
 (a) $\frac{5}{\sqrt{2}}(\hat{i} - \hat{j})$ (b) $\frac{5}{\sqrt{2}}(\hat{i} + \hat{j})$ (c) $5(\hat{i} - \hat{j})$ (d) $5(\hat{i} + \hat{j})$ $\vec{a} = \frac{\hat{i} - \hat{j}}{\sqrt{2}}$
 $|\vec{b}| = \sqrt{3^2 + 4^2} = 5$
- 31) Choose the wrong statement
 (a) Three vectors of different magnitudes may be combined to give zero resultant.
 (b) Two vectors of different magnitude can be combined to give a zero resultant
 (c) The product of a scalar and a vector is a vector quantity.
 (d) All of the above are wrong statements.
- 32) What displacement at an angle 60° to the x-axis has an x-component of 5 m? $\hat{i}$ and $\hat{j}$ are unit vectors in x and y directions, respectively.
 (a) $5\hat{i}$ (b) $5\hat{i} + 5\hat{j}$ (c) $5\hat{i} + 5\sqrt{3}\hat{j}$ (d) All of the above
- 33) Mark the correct statement
 (a) $|\vec{a} + \vec{b}| \geq |\vec{a}| + |\vec{b}|$ (b) $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ (c) $|\vec{a} - \vec{b}| \geq |\vec{a}| + |\vec{b}|$ (d) All of the above
- 34) Out of the following forces, the resultant of which cannot be 10N?
 (a) 15 N and 20 N (b) 10 N and 10 N (c) 5 N and 12 N (d) 12 N and 1 N
- 35) Which of the following pairs of forces cannot be added to give a resultant force of 4 N?
 (a) 2 N and 8 N (b) 2 N and 2 N (c) 2 N and 6 N (d) 2 N and 4 N
- 36) In an equilateral triangle ABC, AL, BM, and CN are medians. Forces along BC and BA represented by them will have a resultant represented by
 (a) 2AL (b) 2BM (c) 2CN (d) AC
- 37) The vector sum of two forces is perpendicular to their vector difference. The forces are
 (a) equal to each other (b) equal to each other in magnitude
 (c) not equal to each other in magnitude (d) cannot be predicted
- 38) If a parallelogram is formed with two sides represented by vector $\vec{a}$ and $\vec{b}$, then $\vec{a} + \vec{b}$ represents the
 (a) major diagonal when the angle between vectors is acute
 (b) minor diagonal when the angle between vectors is obtuse
 (c) both of the above
 (d) none of the above

- 39) The resultant $\vec{C}$ of $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{A}$. Also, $|\vec{A}| = |\vec{C}|$. The angle between $\vec{A}$ and $\vec{B}$ is

(a) $\frac{\pi}{4} \text{ rad}$

(b) $\frac{3\pi}{4} \text{ rad}$

(c) $\frac{5\pi}{4} \text{ rad}$

(d) $\frac{7\pi}{4} \text{ rad}$

- 40) Two forces $\vec{F}_1 = 500 \text{ N}$ due east and $\vec{F}_2 = 250 \text{ N}$ due north have their common initial point. $\vec{F}_2 - \vec{F}_1$ is

(a) $250\sqrt{5} \text{ N}, \tan^{-1}(2) \text{ W of N}$

(b) $250 \text{ N}, \tan^{-1}(2) \text{ W of N}$

(c) zero

(d) $750 \text{ N}, \tan^{-1}(3/4) \text{ N of W}$

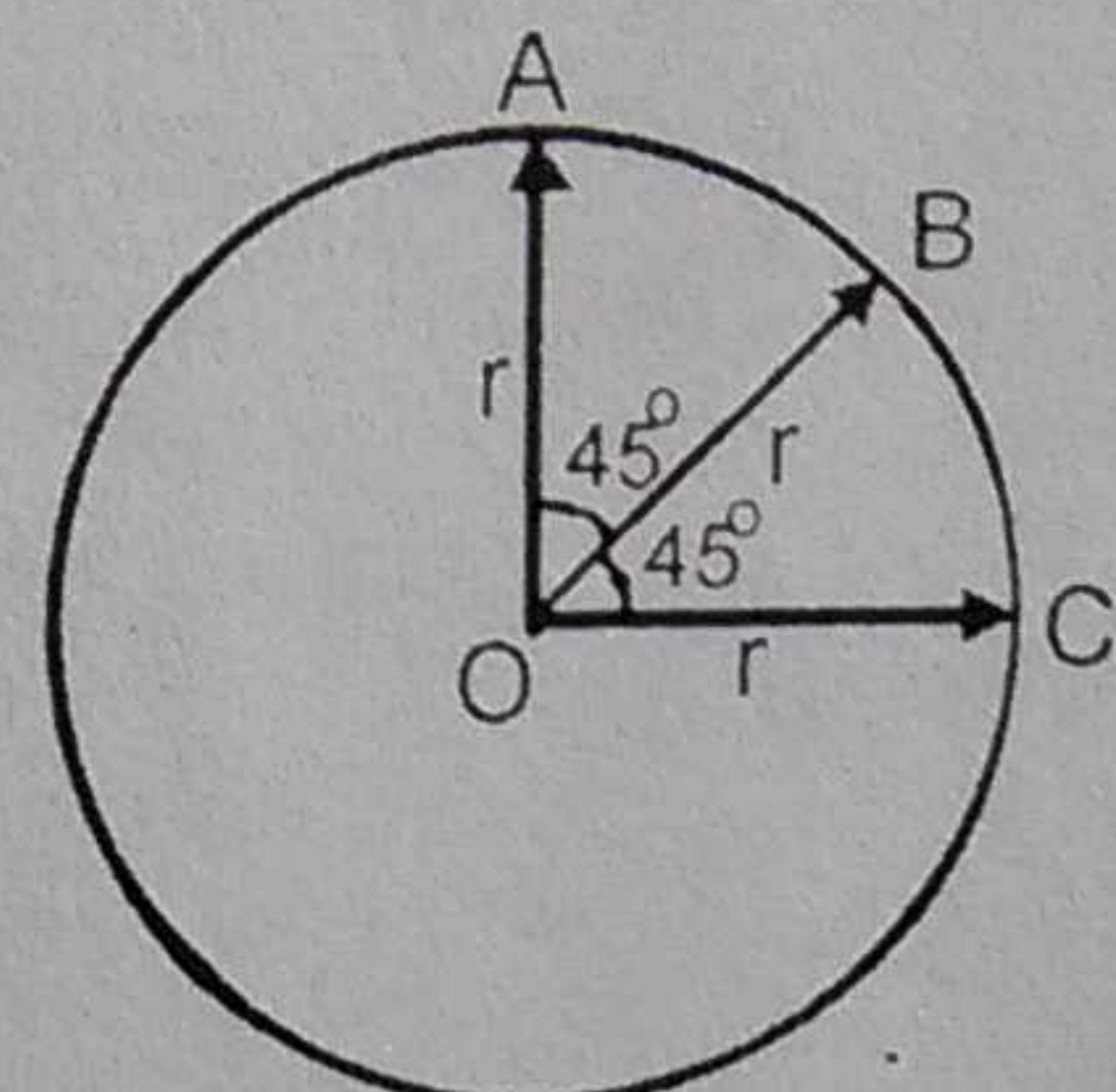
- 41) The resultant of the three vectors $\vec{OA}$, $\vec{OB}$, and $\vec{OC}$ shown in fig. is

(a) r

(b) $2r$

(c) $r(1 + \sqrt{2})$

(d) $r(\sqrt{2} - 1)$



- 42) Two vectors $\vec{a}$ and $\vec{b}$ are at an angle of 60° with each other. Their resultant makes an angle of 45° with $\vec{a}$. If $|\vec{b}| = 2$ units, then $|\vec{a}|$ is

(a) $\sqrt{3}$

(b) $\sqrt{3} - 1$

(c) $\sqrt{3} + 1$

(d) $\sqrt{3}/2$

- 43) The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If the magnitude of $\vec{Q}$ is doubled the new resultant vector becomes perpendicular to $\vec{P}$. Then, the magnitude of $\vec{R}$ is equal to

(a) $P + Q$

(b) P

(c) $P - Q$

(d) Q

Dot product becomes zero

- 44) A vector $\vec{A}$ when added to the vector $\vec{B} = 3\hat{i} + 4\hat{j}$ yields a resultant vector that is in the positive y-direction and has a magnitude equal to that of $\vec{B}$. Find the magnitude of $\vec{A}$

(a) $\sqrt{10}$

(b) 10

(c) 5

(d) $\sqrt{15}$

- 45) ABCDEF is a regular hexagon with point O as centre. The value of $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF}$ is

(a) $2\vec{AO}$

(b) $4\vec{AO}$

(c) $6\vec{AO}$

(d) 0

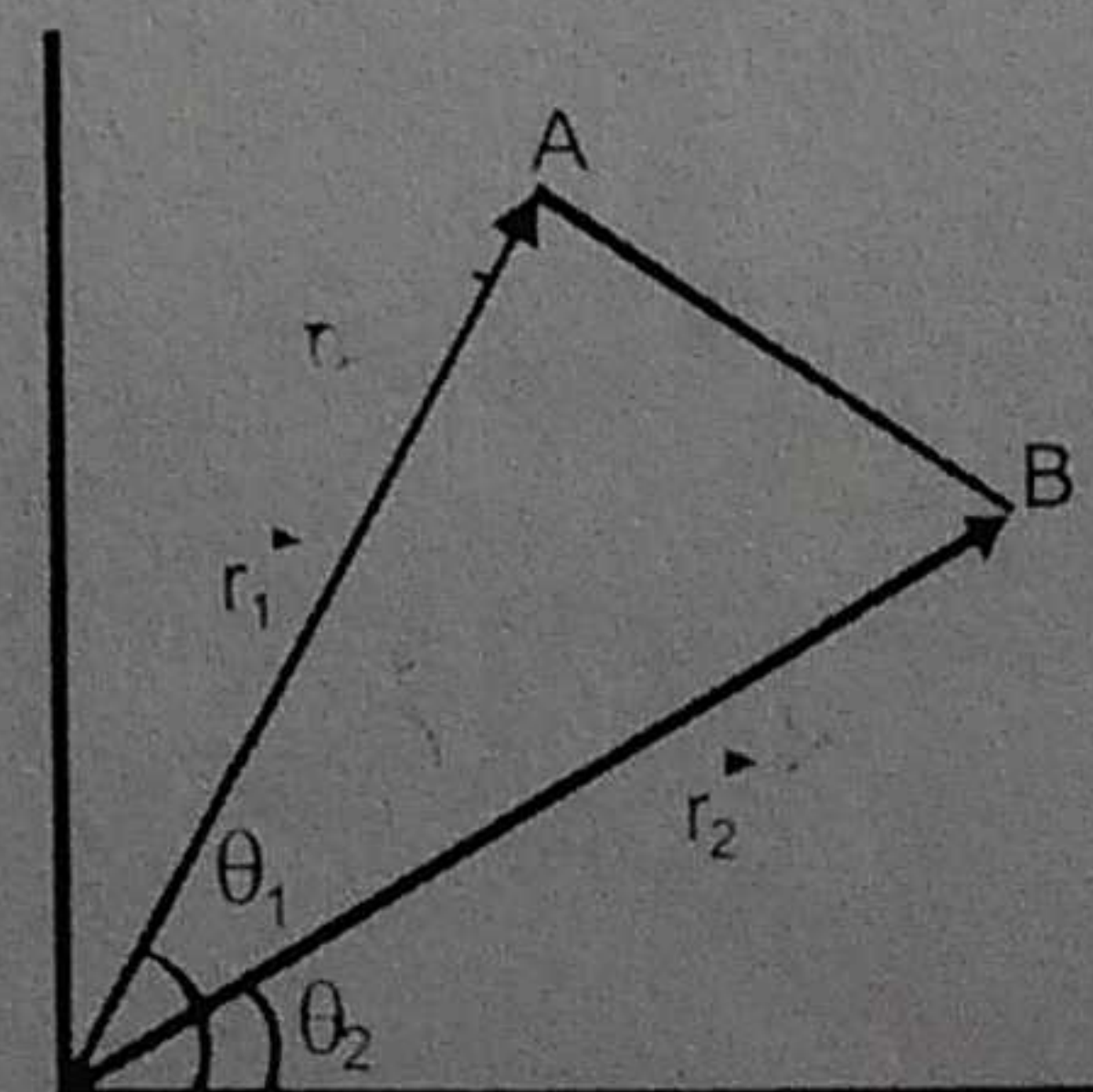
- 46) In a two dimensional motions of a particle, the particle moves from point A, position vector $\vec{r}_1$ to point B, position vector $\vec{r}_2$. If the magnitudes of these vectors are, respectively $r_1 = 3$ and $r_2 = 4$ and the angles they make with the x-axis are $\theta_1 = 75^\circ$ and $\theta_2 = 15^\circ$, respectively, then find the magnitude of the displacement vector.

(a) 15

(b) $\sqrt{13}$

(c) 17

(d) $\sqrt{15}$



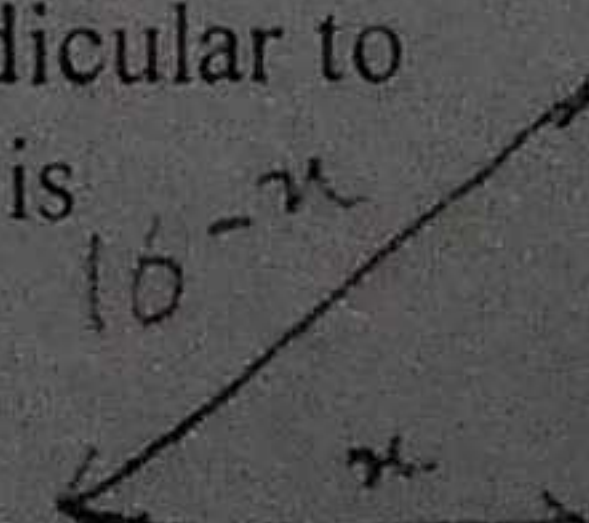
- 47) The sum of the magnitude of two forces acting at a point is 16 N . The resultant of these forces is perpendicular to the smaller force and has a magnitude of 8 N . If the smaller force is of magnitude x , then the value of x is

(a) 2 N

(b) 4 N

(c) 6 N

(d) 7 N



- 48) The angle between two vectors $\vec{A}$ and $\vec{B}$ is θ . Resultant of these vector $\vec{R}$ makes an angle $\theta/2$ with $\vec{A}$. Which of the following is true?

(a) $A = 2B$

(b) $A = B/2$

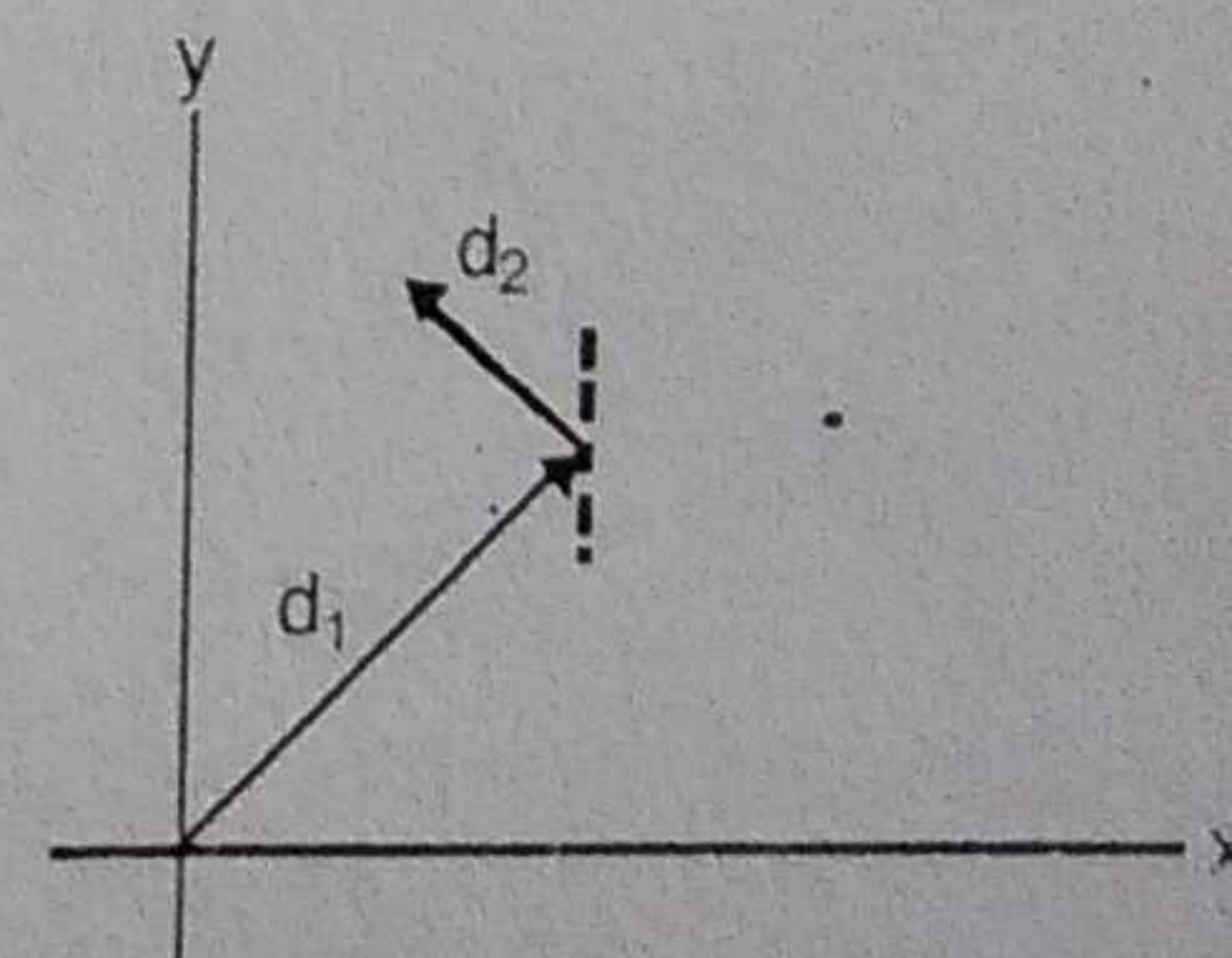
(c) $A = B$

(d) $AB = 1$

- 49) The resultant of three vectors 1, 2, and 3 units whose directions are those of the sides of an equilateral triangle is at an angle of
 (a) 30° with the first vector (b) 15° with the first vector
 (c) 100° with the first vector (d) 150° with the first vector
- 50) The simple sum of two co-initial vectors is 16 units. Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector. The magnitudes of the two vectors are
 (a) 2 units and 14 units (b) 4 units and 12 units (c) 6 units and 10 units (d) 8 units and 8 units
- 51) The components of a vector along x-and y-directions are $(n + 1)$ and 1, respectively. If the coordinate system is rotated by an angle $\theta = 60^\circ$, then the components change to n and 3. The value of n is
 (a) 2 (b) $\cos 60^\circ$ (c) $\sin 60^\circ$ (d) 3.5
- 52) What is the resultant of three coplanar forces: 300 N at 0° , 400 N at 30° , and 400 N at 150° ?
 (a) 500 N (b) 700 N (c) 1,100 N (d) 300 N

Multiple Correct Answers Type

- 53) Which of the following statement is/are correct?
 (a) The signs of x-component of $\vec{d}_1$ is positive and that of $\vec{d}_2$ is negative.
 (b) The signs of the y-component of $\vec{d}_1$ and $\vec{d}_2$ are positive and negative, respectively.
 (c) The signs of x- and y - components of $\vec{d}_1 + \vec{d}_2$ are positive.
 (d) None of these
- 54) Given two vectors $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = \hat{i} + \hat{j}$. θ is the angel between $\vec{A}$ and $\vec{B}$. Which of the following statements is/are correct?
 (a) $|\vec{A}| \cos \theta \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$ is the component of $\vec{A}$ along $\vec{B}$
 (b) $|\vec{A}| \sin \theta \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$ is the component of $\vec{A}$ perpendicular to $\vec{B}$
 (c) $|\vec{A}| \cos \theta \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}} \right)$ is the component of $\vec{A}$ along $\vec{B}$
 (d) $|\vec{A}| \sin \theta \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$ is the component of $\vec{A}$ perpendicular to $\vec{B}$
- 55) If $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ are two vectors then the unit vector
 (a) Perpendicular to $\vec{A}$ is $\left(\frac{-\hat{j} + \hat{k}}{\sqrt{2}} \right)$ (b) Parallel to $\vec{A}$ is $\left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}} \right)$
 (c) Perpendicular to $\vec{A}$ is $\left(\frac{+\hat{j} + \hat{k}}{\sqrt{2}} \right)$ (d) Parallel to $\vec{A}$ is $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- 56) If $(\vec{v}_1 + \vec{v}_2)$ is perpendicular to $(\vec{v}_1 - \vec{v}_2)$, then
 (a) $\vec{v}_1$ is perpendicular to $\vec{v}_2$ (b) $|\vec{v}_1| = |\vec{v}_2|$
 (c) $\vec{v}_1$ is null vector (d) the angle between $\vec{v}_1$ and $\vec{v}_2$ can have any value
- 57) Two vectors $\vec{A}$ and $\vec{B}$ lie in one plane. Vector $\vec{C}$ lies in a different plane. Then $\vec{A} + \vec{B} + \vec{C}$
 (a) cannot be zero
 (b) can be zero
 (c) lies in the plane of $\vec{A}$ or $\vec{B}$
 (d) lies in a plane different from that of any of the three vectors



Key

Single Correct Answer Type

1.a	2.c	3.d	4.d	5.d	6.b	7.b
8.b	9.b	10.a	11.c	12.d	13.a	14.c
15.b	16.d	17.a	18.b	19.c	20.c	21.c
22.b	23.b	24.a	25.a	26.b	27.b	28.b
29.b	30.a	31.b	32.c	33.b	34.d	35.a
36.b	37.b	38.c	39.b	40.a	41.c	42.b
43.d	44.a	45.c	46.b	47.c	48.c	49.a
50.c	51.d	52.a				

Multiple Correct Answers Type

53.a,c	54.a,b	55.a,b,c	56.a,b	57.ad
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SUBJECTIVE (VECTORS)

1. Physical quantities which have only magnitude are called scalars. Physical quantities which have magnitude as well as direction and obey laws of vector add. are called vectors.

2. We know that parallel translation of a vector does not bring about any change in it. Let us represent a vector as line segment in xy plane with head and tail as coordinates of its end points. If $\vec{A}$ is represented as:

Head: (4, 7)

Tail: (2, 3)

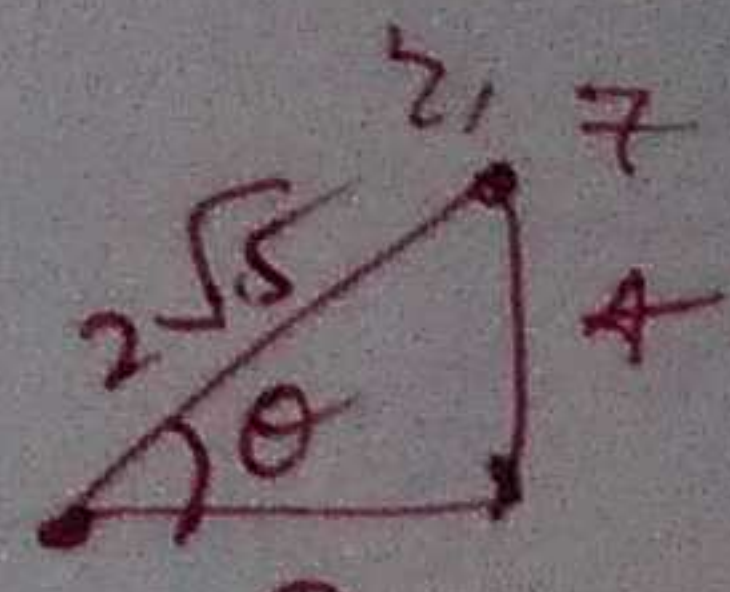
Then what is the general form of line segments (in head tail format) representing all vectors $(0,0)$.

parallel to $\vec{A}$.

$$4(3-y) + 2(y-7) + x(4) = 0$$

$$12 - 4y + 2y - 14 + 4x = 0$$

$$4x - 2y - 2 = 0$$

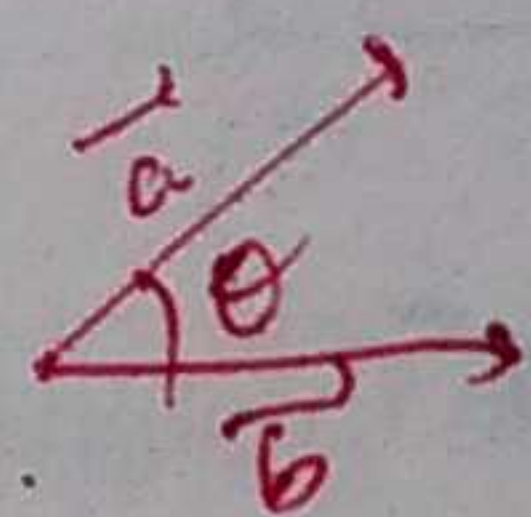


$$\tan \theta = 2$$

$$2\sqrt{5} = \sqrt{x^2 + y^2}$$

$$x^2 + y^2 = 2\sqrt{5}$$

3. Two vectors having equal magnitude A make an angle θ with each other. Find the magnitude and direction of the resultant.



$$\sqrt{2a^2 + 2a^2 \cos \theta}$$

$$\sqrt{2a^2 (1 + \cos \theta)}$$

4. $k_1 \vec{a} + k_2 \vec{b} + x \vec{a} + y \vec{b} = 0$

If k_1 and k_2 are constants, x and y are unknown numbers, what are possible values of x and y in two cases

i) If $\vec{a}$ and $\vec{b}$ are not parallel

ii) If $\vec{a}$ and $\vec{b}$ are parallel

5. Two vectors of equal magnitude 5 unit have an angle of 60° between them. Find the magnitude of

i) The sum of the vectors.

ii) The difference of the vectors.

6. $\vec{a}$ lying completely in xy plane has magnitude M and makes an angle α with the x-axis. $\vec{b}$ has

same magnitude and makes an angle α with y-axis. Find out the resultant of two vectors using vector resolution. Does the direction of resultant depend on α ?

7. Write the unit vector in the direction of $\vec{A} = 5\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

8. Magnitudes of 3 vectors are given in sets. Which of the following sets can give zero resultant?

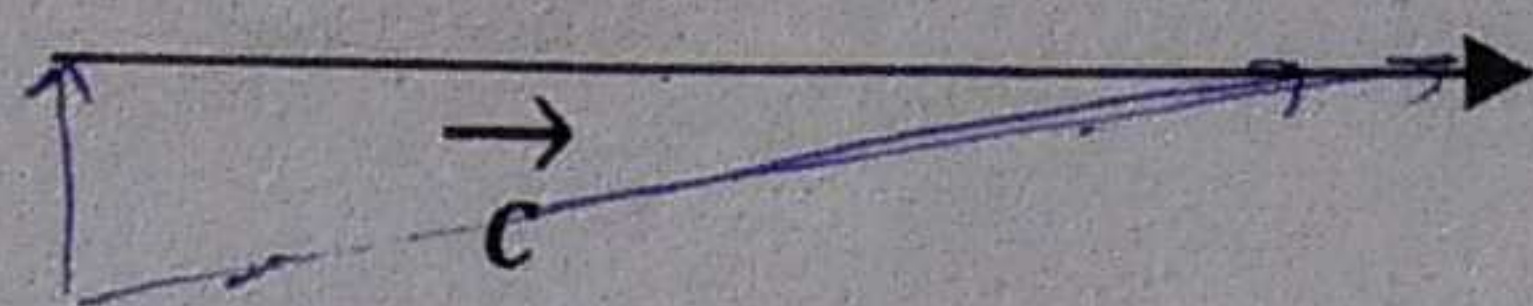
i) 2, 4, 8

ii) 9, 8, 16

iii) 1, 2, 1

iv) 0.5, 1, 2

9. Resultant $\vec{c}$ of two perpendicular vectors (vectors $\vec{a}$ and $\vec{b}$) is shown. Given that $|\vec{a}| = 5$ and $|\vec{c}| = 10$. Draw $\vec{a}$ and $\vec{b}$.

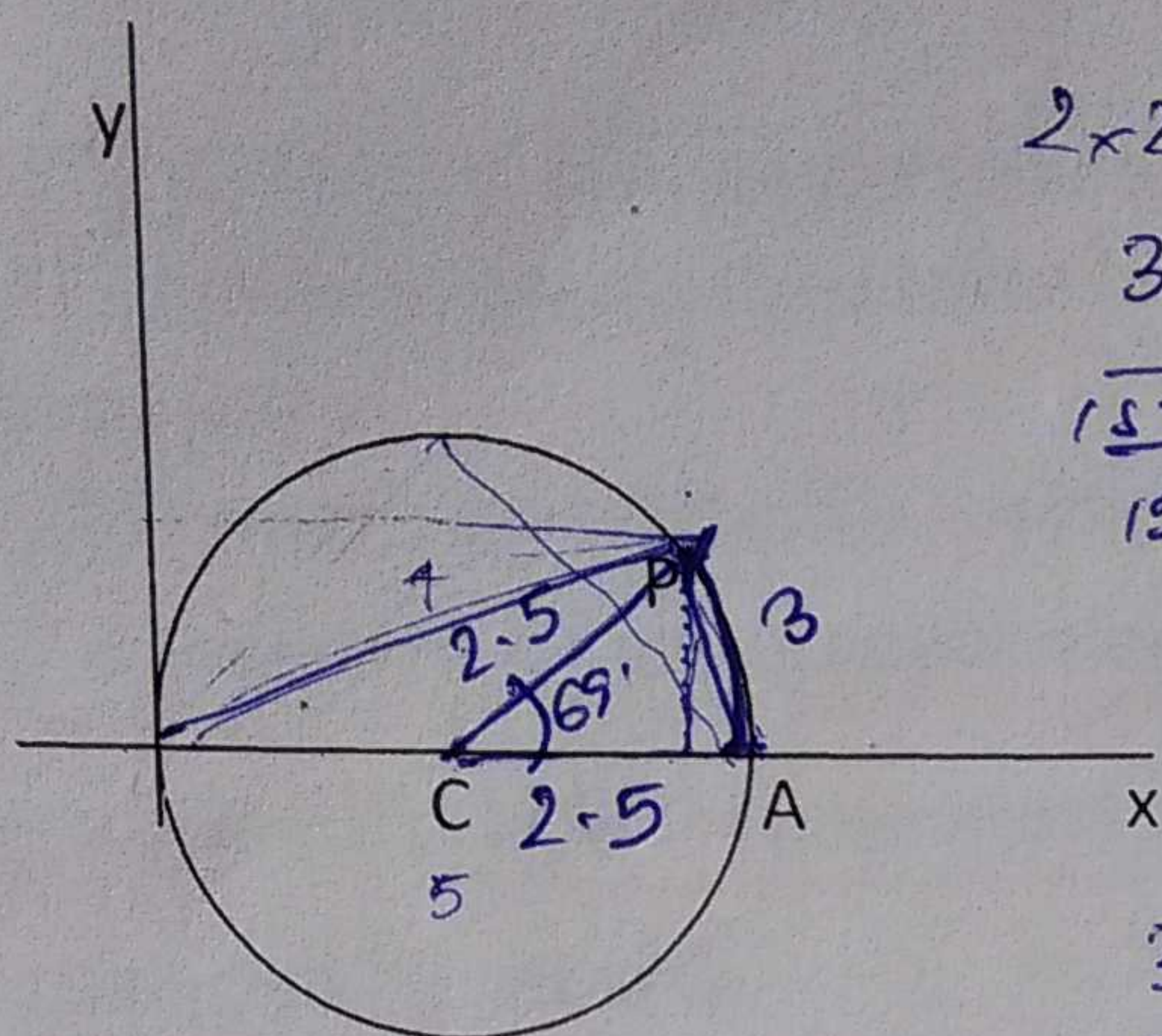


$$10 = \sqrt{25 + 25 + 2 \times 5 \times 5 \cos \theta}$$

$$100 = 50 + 50 \cos \theta$$

$$\cos \theta = 1 \quad \theta = 0$$

10. A circle with its centre C lying on x-axis and passing through origin cuts x-axis at point A. Radius of circle is 2.5cm and AP=3cm. What is position vector of point P?



$$\pi : 180$$

$$2 \times 2.5 \times 3.14$$

$$3.14$$

$$\frac{15.6}{5}$$

$$15.6 = 2\pi r$$

$$\frac{156}{2} = 78$$

$$\frac{156}{3} = 52$$

$$\frac{156}{4} = 39$$

$$\frac{156}{6} = 26$$

$$\frac{156}{12} = 13$$

$$3 = \frac{\theta}{360} \times 2 \times 2.5 \times 3.14$$

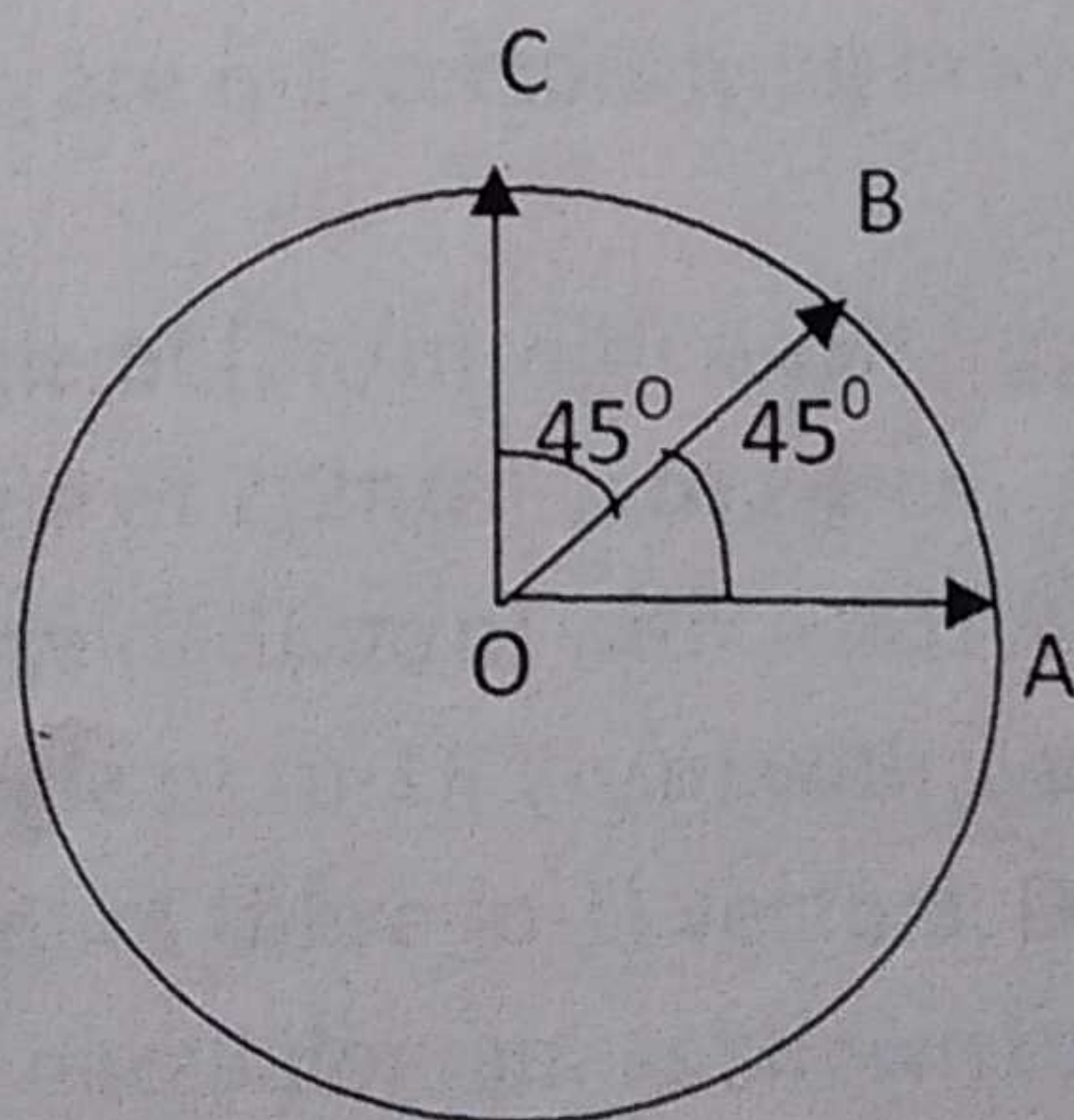
$$\frac{3 \times 360}{15.6} = \theta \quad 2.3 \times 10 \times 3 = \theta$$

$$69' = \theta$$

11. The resultant of $\vec{a}$ and $\vec{b}$ makes an angle of α with $\vec{a}$ and β with $\vec{b}$. When is

- $\alpha > \beta$
- $\beta > \alpha$
- $\alpha = \beta$

12. Find the resultant of the three vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ shown in figure. Radius of the circle is R.



$$|\vec{OA}| = r$$

$$\therefore \sqrt{r^2 + r^2 + 2 \times r \times r \cos 90}$$

$$\sqrt{r^2 + r^2}$$

$$\sqrt{2r^2}$$

$$\sqrt{2} r = \vec{OB}$$

$$\sqrt{2} r + r = (\sqrt{2} + 1)r$$

13. Is a vector necessarily changed if it is rotated through an angle? (Is it possible to add two vectors of unequal magnitudes and get zero? Is it possible to add 3 vectors of equal magnitudes and get zero?) Yes

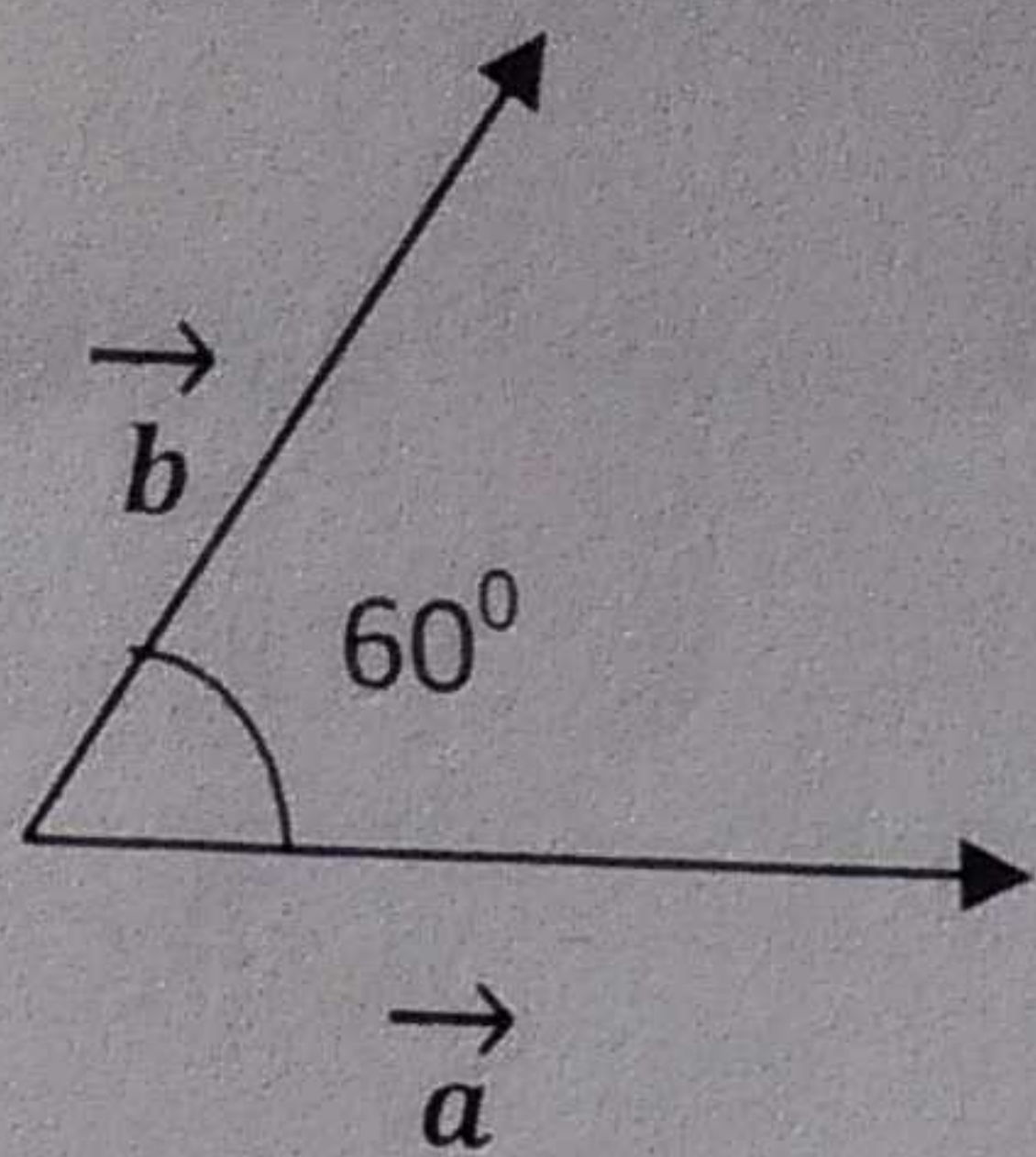
14. Let $\vec{a} = 4i + 3j$ and $\vec{b} = 3i + 4j$. Find the magnitude of $\vec{a}$, $\vec{b}$, $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$

$$5 \quad 5 \quad 7\sqrt{2} \quad \sqrt{2}$$

15. $\vec{a}$ and $\vec{b}$ are as shown in figure. A is kept constant while magnitude of b is varied keeping its direction unchanged. If $0 < |\vec{b}| < \infty$, how will direction of resultant vary with $|\vec{b}|$?

$$0 = a^2 + b^2 + 2ab \cos \theta$$

$$= (-1)(a^2 + b^2)$$



16. Let $A_1 A_2 A_3 A_4 A_5 A_6 A_1$ be a regular hexagon. Write the x-components of the vectors represented by the six sides taken in order. Use the fact that the resultant of six vectors is zero, to prove that

$$\cos(0) + \cos(\pi/3) + \cos(2\pi/3) + \cos(3\pi/3) + \cos(4\pi/3) + \cos(5\pi/3) = 0.$$

17. $\vec{a}$ is such that its component along x-axis and y-axis are equal and its component along z-axis is $\sqrt{2}$ times its component along x-axis. Find the angle which $\vec{a}$ makes with the three coordinate axis. $(x, x, \sqrt{2}x) \quad (x\hat{i} + x\hat{j} + \sqrt{2}x\hat{k})$

18. $\vec{a}$ lying completely in xy plane has magnitude a and makes an angle α with y-axis. The entire coordinate system is now rotated by angle of θ in anti-clockwise sense. What are the components of vector along the new axis? Instead of resolving $\vec{a}$ along the new axis if we resolve the old components of $\vec{a}$ along new axis will we get the same result? If yes, show how?

19. If $\vec{a}$ makes equal angle with the three coordinate axis, what is the unit vector in the direction of $\vec{a}$?

20. $A_1, A_2, A_3, \dots, A_{n-1}$ are $n-1$ vectors lying in xy plane each of magnitude A. (n is positive integer such that $n \geq 2$)

A_1 makes an angle of $(2\pi/n)$ with x-axis

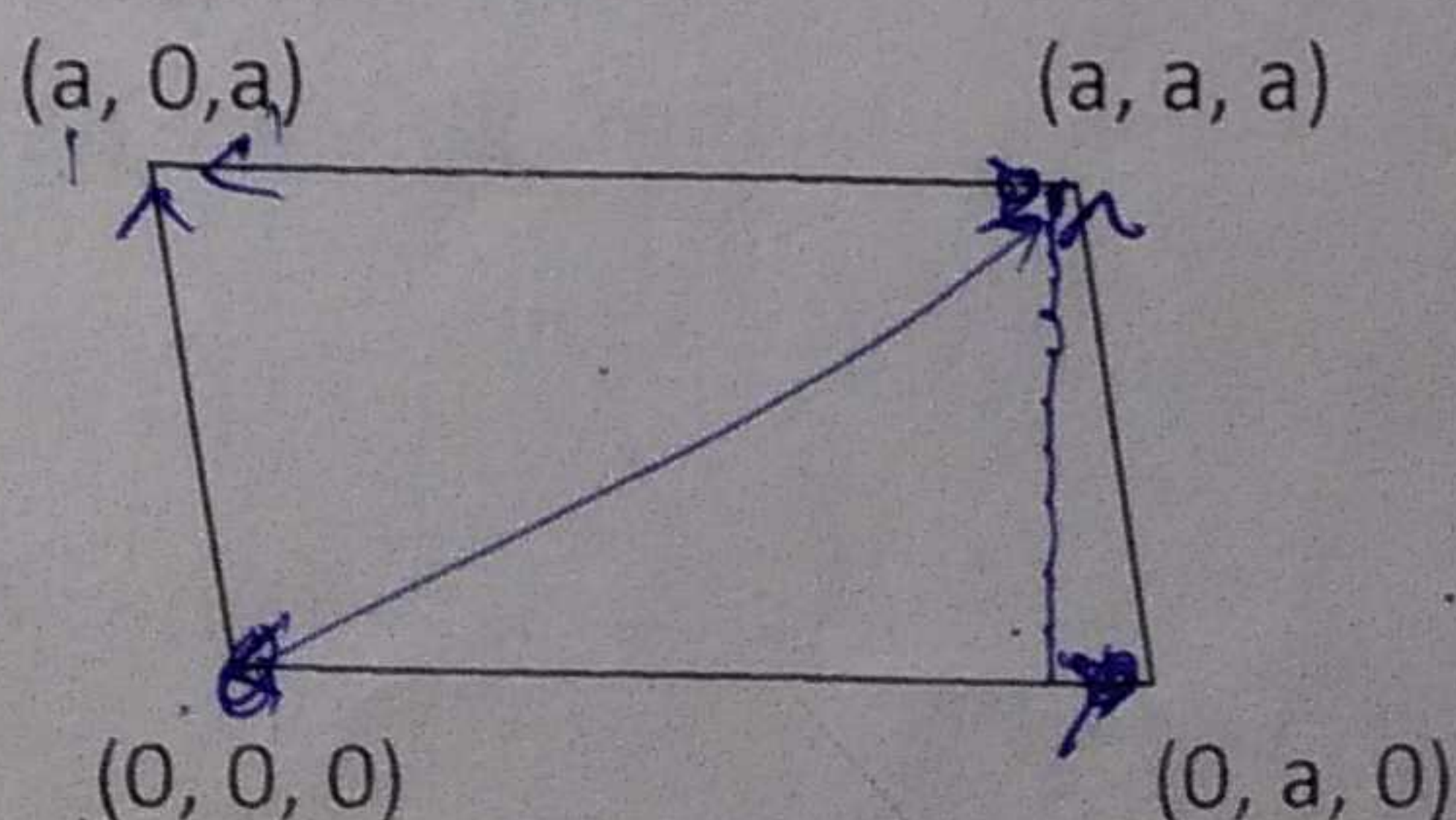
A_2 makes an angle of $2(2\pi/n)$ with x-axis

A_3 makes an angle of $3(2\pi/n)$ with x-axis

$\dots A_{n-1}$ makes angle of $(n-1)(2\pi/n)$ with x-axis

Find the resultant of these $(n-1)$ vectors. Does it depend on value of n?

21. Find area of the parallelogram as shown using vector product



(a, a, a)

$(a\hat{i} + a\hat{j})$

$a\hat{j}$

$\vec{a} \times \vec{b}$

$ab \sin \theta$

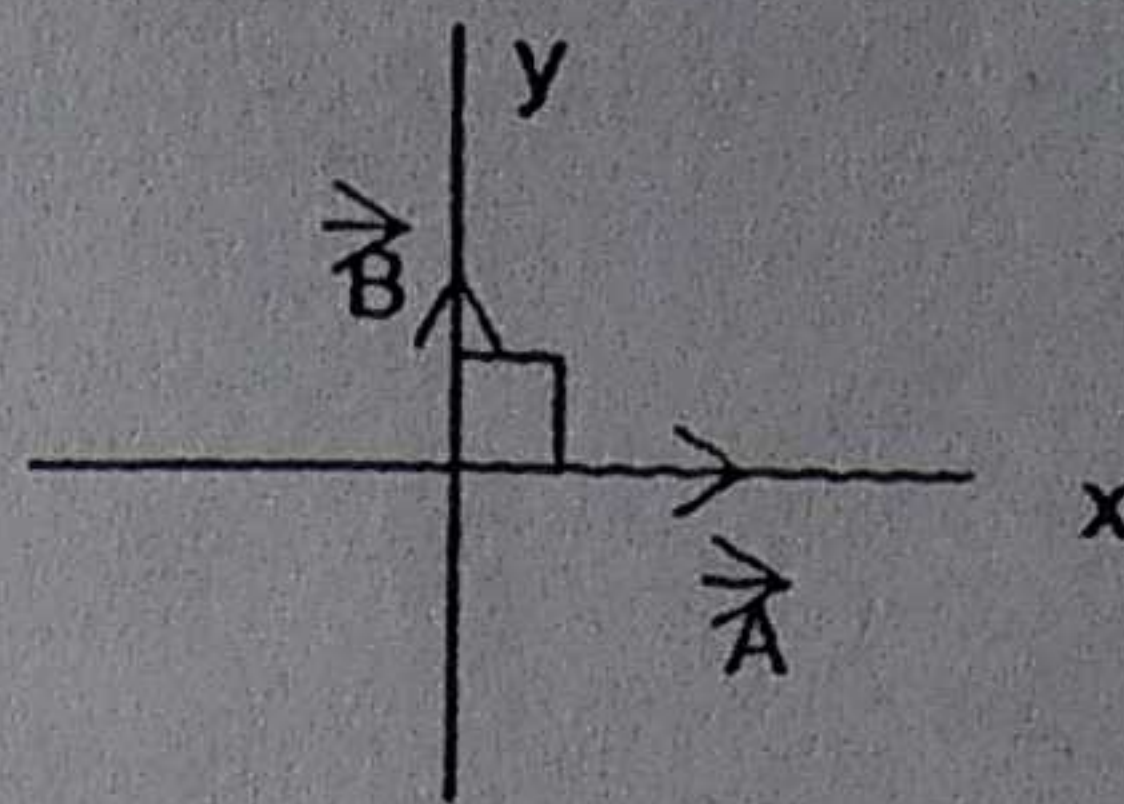
$ab |\vec{a} + \vec{b}| = \sqrt{a^2 + b^2} = \sqrt{2}a$

$(a\hat{j})$

a

$\therefore \sqrt{2}a \times a = \sqrt{2}a^2$

VECTOR*



- ✓ 1. $\vec{A}$ and $\vec{B}$ are shown in the figure. where $|\vec{A}| = |\vec{B}| = 5$ units. $\vec{C}$ is a vector such that $\vec{A} + \vec{B} + \vec{C} = 0$. What is the magnitude of $\vec{C}$?
- ✓ 2. If $\vec{A} = 2\hat{i} - 3\hat{j}$ and $\vec{B} = 2\hat{k}$, then what is the value of $\vec{A} \cdot \vec{B}$.
- ✓ 3. For the resultant of two vectors to be maximum what must be the angle between them
- ✓ 4. Find the angle between vectors $\vec{A} = 4\hat{i} + 3\hat{j}$ & $\vec{B} = 3\hat{i} - 4\hat{j}$
- ✓ 5. If $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$ then, find the angle between vectors $\vec{A}$ and $\vec{B}$
- ✓ 6. If $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} + 2\hat{k}$, find unit vector which is perpendicular to both $\vec{A}$ and $\vec{B}$.
- ✓ 7. Two forces F_1 and F_2 are acting on a particle and the angle between them is θ . The angle between the resultant force and F_1 is α . Find $\tan \alpha$ in terms of F_1 , F_2 and θ .
- ✓ 8. Three force $\vec{F}_1 = (3\hat{i} + 2\hat{j} - \hat{k})\text{N}$; $\vec{F}_2 = (3\hat{i} + 4\hat{j} - 5\hat{k})\text{N}$ and $\vec{F}_3 = A(\hat{i} + \hat{j} - \hat{k})\text{N}$ act simultaneously on a particle. In order that the particle remains in equilibrium, what is the value of A.
- ✓ 9. If vectors $\vec{A}$ & $\vec{B}$ have an angle θ between them then, what is the value of $\frac{|\hat{A} - \hat{B}|}{|\hat{A} + \hat{B}|}$
- ✓ 10. If $\vec{A} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ & $\vec{B} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$. Then what is the component of $\vec{B} - \vec{A}$ along z-axis.
- ✓ 11. The resultant of two equal forces is double of either of force. What is the angle between them.
- ✓ 12. If the magnitude of two vectors are 3 and 4 respectively. Prove that the resultant of these 2 vectors cannot exceed 7 units.
- ✓ 13. Vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = -\hat{i} - \hat{j} - \hat{k}$, what angle will $(\vec{A} - \vec{B})$ make with $\vec{A}$
- ✓ 14. Given that $\vec{A} + \vec{B} + \vec{C} = 0$, out of three vectors two are equal in magnitude and the magnitude of third vectors is $\sqrt{2}$ times that of either of two having equal magnitude. Find the angle between the vectors.
- ✓ 15. A particle undergoes three successive displacements given by $\vec{s}_1 = \sqrt{2}$ m North-East $\vec{s}_2 = 2\text{m}$ due south and $\vec{s}_3 = 4\text{m}$, 30° north of west, then find magnitude of net displacement.

CPP

VECTOR*

ANSWERS

1. $5\sqrt{2}$

2. Zero.

3. 0°

4. 90°

5. 0°

6. $\frac{3\hat{i} - 4\hat{j} - \hat{k}}{\sqrt{26}}$

7. $\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta}$

8. -6

9. $\tan \theta/2$

10. $b_2 - a_2$

11. 0°

12. 9

13. 0°

14. 90° and 135°

MCQ SINGLE CORRECT (+3, -1) Total = 24

- 1) If $\vec{A} = \vec{B} + \vec{C}$ and the magnitude of $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 5, 4 and 3 units respectively the angle between $\vec{A}$ and $\vec{B}$ is

a) $\cos^{-1}\left(\frac{3}{5}\right)$ b) $\cos^{-1}\left(\frac{4}{5}\right)$ c) $\left(\frac{\pi}{2}\right)$ d) $\cos^{-1}\left(\frac{3}{4}\right)$

- 2) What is the component of $3\hat{i} + 4\hat{j}$ along $\hat{i} + \hat{j}$:

a) $\frac{1}{2}(\hat{i} + \hat{j})$ b) $\frac{3}{2}(\hat{i} + \hat{j})$ c) $\frac{5}{2}(\hat{i} + \hat{j})$ d) $\frac{7}{2}(\hat{i} + \hat{j})$

- 3) If a vector $\vec{A}$ makes an angle α , β and γ with X, Y and Z axis respectively then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \dots\dots\dots$

a) 0 b) 1 c) 2 d) 3

- 4) The X and Y-component of $\vec{P}$ are $7\hat{i}$ and $6\hat{j}$. Also, the X and Y-components of $\vec{P} + \vec{Q}$ are $11\hat{i}$ and $9\hat{j}$ respectively. The magnitude of $\vec{Q}$ is:

a) 7 b) 6 c) 5 d) 13

- 5) Two vector $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{B} = 4\hat{i} + 8\hat{j} + x\hat{k}$ are such that the component of $\vec{B}$ along $\vec{A}$ is zero. Then the value of x will be

a) 8 b) -4 c) +4 d) -8

- 6) A blind person after walking 10 steps in one direction turns randomly to left or right. After walking 'n' steps, his displacement is $16\sqrt{2}$ m. Assuming length of each step to be 80 cm, find the minimum value of 'n'.

a) 20 b) 30 c) 40 d) 60

- 7) If the resultant of two vectors having magnitudes of 7 and 4 is 3, then the magnitude of the cross product of the two vectors will be:

a) 28 b) $\sqrt{65}$ c) $\sqrt{33}$ d) Zero

- 8) The adjacent of sides of a parallelogram is represented by vectors $2\hat{i} + 3\hat{j}$ and $\hat{i} + 4\hat{j}$. The area of the parallelogram is:

a) 5 units b) 3 units c) 8 units d) 11 units

MCQ MULTICORRECT (+4, 0) Total = 8

- 1) Which of the following relations are not commutative

a) $\vec{A} - \vec{B}$ b) $\vec{A} + \vec{B}$ c) $\vec{A} \times \vec{B}$ d) $\vec{A} \cdot \vec{B}$

- 2) Which of the following expression is equal to zero? Assuming that $\vec{A}$ and $\vec{B}$ are non parallel vectors.

a) $\vec{A} \times \vec{A}$ b) $\vec{A} \cdot (\vec{A} \times \vec{B})$ c) $\vec{A} \times \vec{B}$ d) $\hat{i} \times (\hat{j} \times \hat{k})$

LINKED COMPERHENSION TYPE (+3, -1) Total = 12

$\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are 4 non-zero vectors. Such that

$$\vec{a} \times \vec{b} = \vec{c}$$

$$\vec{c} \cdot \vec{d} = 0$$

$$|\vec{a}| = 2, |\vec{b}| = 3$$

$$|\vec{c}| = 3, |\vec{d}| = 2$$

$$ab \sin \theta = c \Rightarrow 6 \sin \theta = 3$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$(\hat{i} + \hat{j}) \cdot \hat{i} + \hat{j} = 3 + 4 = 7$$

A

$$7 = 5 \times \sqrt{2} \cos \theta$$

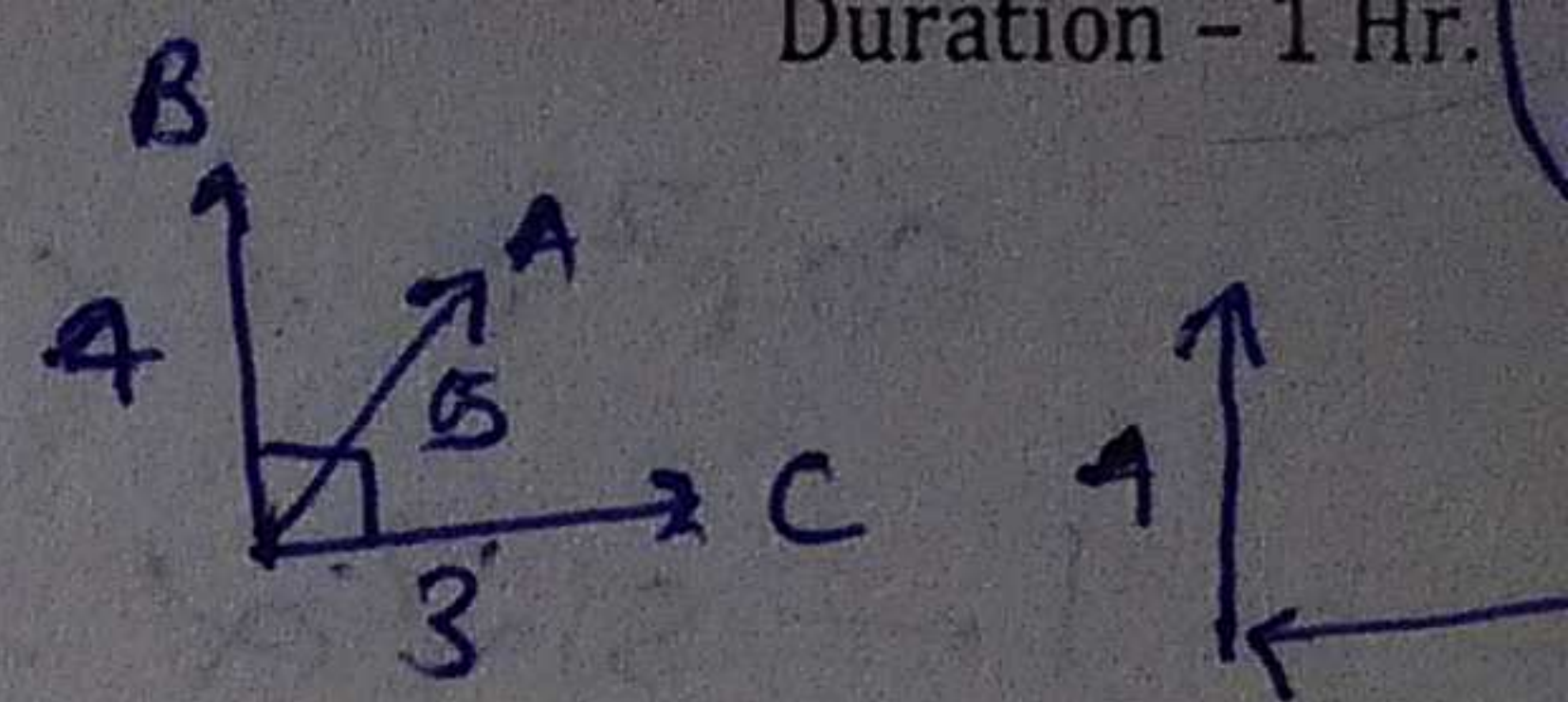
$$A + B = B + A$$

$$a \cdot b = ab \cos \theta$$

$$a \cdot b = 4 \times 3 \times \cos 30^\circ$$

$$a \cdot b = 12 \times \frac{\sqrt{3}}{2}$$

$$a \cdot b = 6\sqrt{3}$$



Find the value of following

1. $\vec{a} \cdot \vec{b}$
 a) 0 b) $3\sqrt{3}$ c) $3\sqrt{2}$ d) $2\sqrt{3}$
2. $\vec{b} \cdot \vec{c}$
 a) 0 b) 2 c) 3 d) 6
3. $|\vec{b} \times \vec{c}|$
 a) 0 b) $\frac{9\sqrt{3}}{2}$ c) $\frac{9}{2}$ d) 9
4. $|\vec{c} \times \vec{d}|$
 a) 0 b) $2\sqrt{3}$ c) $3\sqrt{2}$ d) 6

MATRIX MATCH (+3, 0) Total = 12

⇒ Match the following

	Table - I	Table - II
(q, r)	(a) Speed	(p) Vector
(p, t)	(b) $\hat{i}(0)$	(q) Scalar (a)
(p, s)	(c) $\hat{i} + \hat{j} + \hat{k}$	(r) Makes angle of 0° with positive direction (a)
(p, s)	(d) $\hat{i} - \hat{j} + \hat{k}$	(s) magnitude $\sqrt{3}$ (c)
		(t) null vector (b)

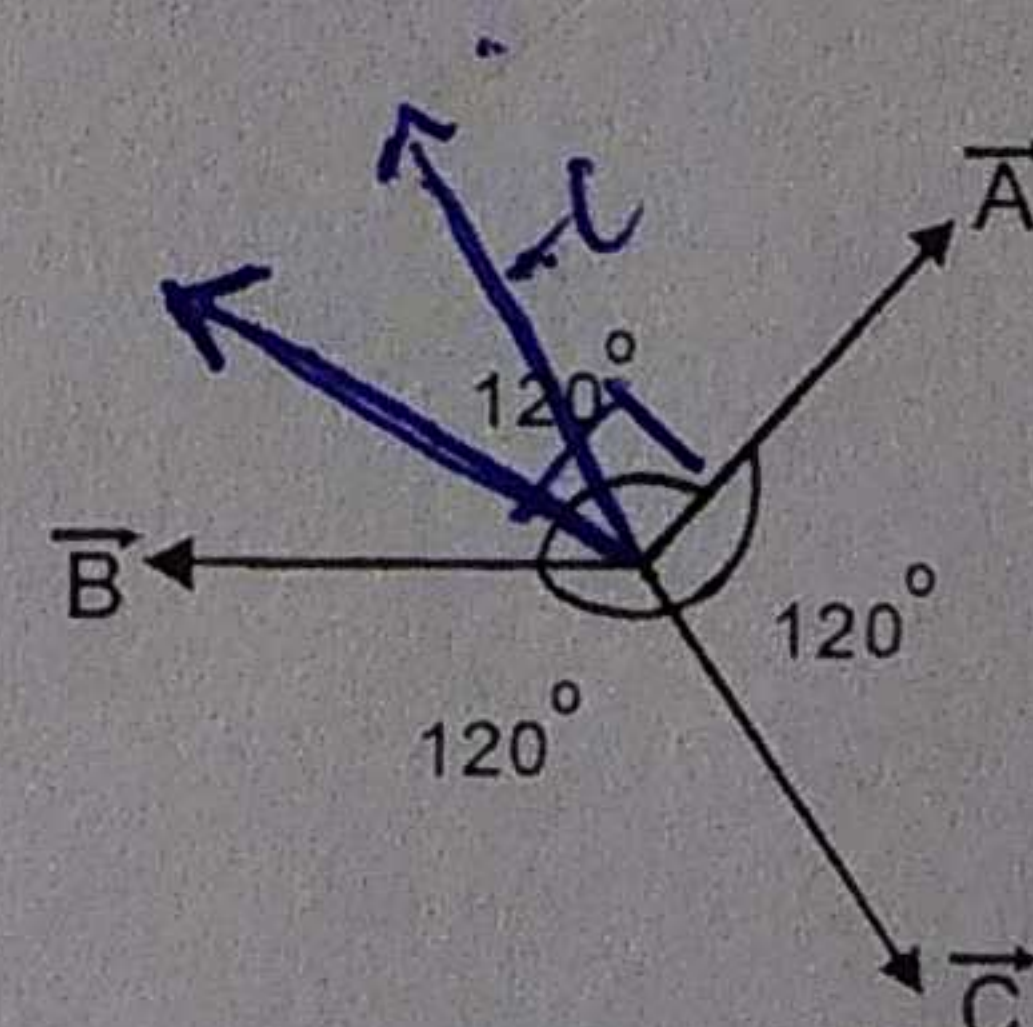
INTEGER TYPE (+4, 0) Total = 12

1. Considering two vectors $\vec{F} = (4\hat{i} - 10\hat{j})N$ and $\vec{r} = (-5\hat{i} - 3\hat{j})m$, compute $\frac{|\vec{r} \times \vec{F}|}{31}$? 2
2. Vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ have magnitudes 5, 12 and 13 units and $\vec{P} + \vec{Q} = \vec{R}$. If angle between $\vec{Q}$ and $\vec{R}$ is θ . Then find $\frac{13}{2} \cos \theta$. 6
3. If $\vec{A}$ and $\vec{B}$ are two mutually perpendicular vectors, where $\vec{A} = 5\hat{i} + 7\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} + 2\hat{j} - a\hat{k}$, then find the value of a . 8

SUBJECTIVE (+6, 0) Total = 12 (Partial Marking Applicable)

1. Three vectors $\vec{A}, \vec{B}$ and $\vec{C}$ have the same magnitude A. find magnitude of following vectors.

- a) $\vec{B} + \vec{C}$ $|\vec{A}| = |\vec{B}| = |\vec{C}|$
 b) $\vec{A} + \vec{B} + \vec{C}$ 0
 c) $\vec{B} - \vec{C}$ $3|\vec{A}|$
 d) $\vec{A} + \vec{B} - \vec{C}$ $2|\vec{A}|$



1. $\vec{a}$ is such that its component along x-axis and y-axis are equal and its component along z-axis is $\sqrt{2}$ times its component along x-axis. Find the angle which $\vec{a}$ makes with the three coordinate axis.

$60^\circ, 60^\circ, 45^\circ$

QUIZ – PHYSICS (VECTORS)

Answer Key

MCQ SINGLE CORRECT

✓ 1.b ✓ 2.d ✓ 3.c ✓ 4.c ✓ 5.a ✗ 6.c ✓ 7.d ✓ 8.a

MCQ MULTICORRECT

✓ 1. a, c ✓ 2. a, b, d

LINKED COMPERHENSION TYPE

✓ 1.b ✓ 2.a ✓ 3.d ✓ 4.d

MATRIX MATCH

✗ A – q, ✓ B – p, t ✓ C – p, s ✓ D – p, s

INTEGER TYPE

✓ 1. 2 ✓ 2. 6 ✓ 3. 8

SUBJECTIVE

✓ 1. $A, 0, \sqrt{3}A, 2A$ ✓ 2. $60^\circ, 60^\circ, 45^\circ$

73
—
80